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August 2016

# FAN7842
# High- and Low-Side Gate Driver

## Features

- Floating Channels Designed for Bootstrap Operation to +200 V
- Typically 350 mA/650 mA Sourcing/Sinking Current Driving Capability for Both Channels
- Common-Mode dv/dt Noise Canceling Circuit
- Extended Allowable Negative $V_S$ Swing to -9.8 V for Signal Propagation at $V_{CC}$=$V_{BS}$=15 V
- $V_{CC}$ & $V_{BS}$ Supply Range from 10 V to 20 V
- UVLO Functions for Both Channels
- TTL Compatible Input Logic Threshold Levels
- Matched Propagation Delay Below 50 nsec
- Output In-phase with Input Signal

## Applications

- Battery based motor applications (E-bike, Power Tool)
- Telecom DC-DC converter

## Related Resources

- *AN-6076 - Design and Application Guide of Bootstrap Circuit for High-Voltage Gate-Drive IC*
- *AN-9052 - Design Guide for Selection of Bootstrap Components*
- *AN-8102 - Recommendations to Avoid Short Pulse Width Issues in HVIC Gate Driver Applications*

## Description

The FAN7842, a monolithic high and low side gate drive IC, which can drive MOSFETs and IGBTs that operate up to +200 V.

Fairchild's high-voltage process and common-mode noise canceling technique provide stable operation of the high-side driver under high-dv/dt noise circumstances. An advanced level-shift circuit allows high-side gate driver operation up to $V_S$=-9.8V (typical) for $V_{BS}$=15V. The input logic level is compatible with standard TTL-series logic gates.

The UVLO circuits for both channels prevent malfunction when $V_{CC}$ and $V_{BS}$ are lower than the specified threshold voltage. Output driver current (source/sink) is typically 350mA/650mA, respectively.

**8-SOP**



## Ordering Information

| Part Number | Package | Operating Temperature Range | Packing Method |
|---|---|---|---|
| FAN7842MX(1) | 8-SOP | -40°C ~ 125°C | Tape & Reel |

**Note:**

1. These devices passed wave soldering test by JESD22A-111.

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## Typical Application Circuit

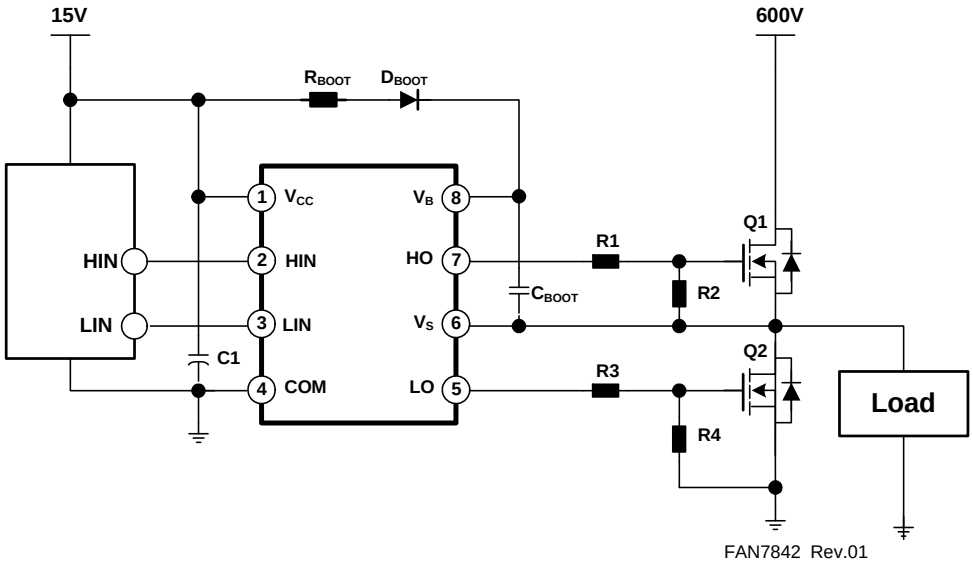



**Figure 1. Application Circuit for Half-Bridge**

## Internal Block Diagram

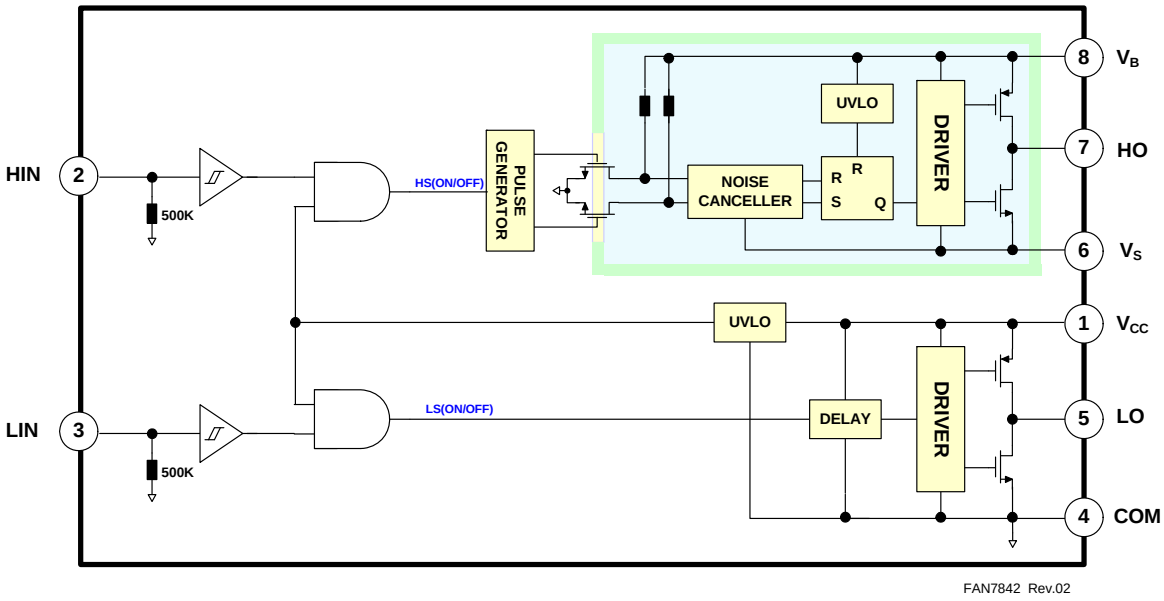



**Figure 2. Functional Block Diagram**

## Pin Assignments

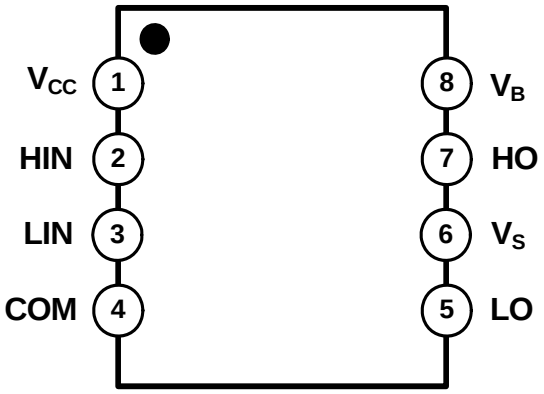



**Figure 3. Pin Configuration (Top View)**

## Pin Definitions

| Name | Description |
|---|---|
| $V_{CC}$ | Low-Side Supply Voltage |
| HIN | Logic Input for High-Side Gate Driver Output |
| LIN | Logic Input for Low-Side Gate Driver Output |
| COM | Logic Ground and Low-Side Driver Return |
| LO | Low-Side Driver Output |
| $V_S$ | High-Voltage Floating Supply Return |
| HO | High-Side Driver Output |
| $V_B$ | High-Side Floating Supply |

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## Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

| Symbol | Characteristics | Min. | Max. | Unit |
|---|---|---|---|---|
| $V_S$ | High-side offset voltage | $V_B$-25 | $V_B$+0.3 | V |
| $V_B$ | High-side floating supply voltage | -0.3 | 225 | |
| $V_{HO}$ | High-side floating output voltage HO | $V_S$-0.3 | $V_B$+0.3 | |
| $V_{CC}$ | Low-side and logic fixed supply voltage | -0.3 | 25 | |
| $V_{LO}$ | Low-side output voltage LO | -0.3 | $V_{CC}$+0.3 | |
| $V_{IN}$ | Logic input voltage (HIN, LIN) | -0.3 | $V_{CC}$+0.3 | |
| COM | Logic ground | $V_{CC}$-25 | $V_{CC}$+0.3 | |
| $dV_S/dt$ | Allowable offset voltage slew rate | | 50 | V/ns |
| $P_D$[(2)(3)(4)] | Power dissipation | | 0.625 | W |
| $\theta_{JA}$ | Thermal resistance, junction-to-ambient | | 200 | °C/W |
| $T_J$ | Junction temperature | | 150 | °C |
| $T_{STG}$ | Storage temperature | | 150 | °C |

**Notes:**

2. Mounted on 76.2 x 114.3 x 1.6 mm PCB (FR-4 glass epoxy material).
3. Refer to the following standards:
   JESD51-2: Integral circuits thermal test method environmental conditions - natural convection
   JESD51-3: Low effective thermal conductivity test board for leaded surface mount packages
4. Do not exceed $P_D$ under any circumstances.

## Recommended Operating Ratings

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. Fairchild does not recommend exceeding them or designing to Absolute Maximum Ratings.

| Symbol | Parameter | Min. | Max. | Unit |
|---|---|---|---|---|
| $V_B$ | High-side floating supply voltage | $V_S$+10 | $V_S$+20 | V |
| $V_S$ | High-side floating supply offset voltage | 6-$V_{CC}$ | 200 | |
| $V_{HO}$ | High-side (HO) output voltage | $V_S$ | $V_B$ | |
| $V_{LO}$ | Low-side (LO) output voltage | COM | $V_{CC}$ | |
| $V_{IN}$ | Logic input voltage (HIN, LIN) | COM | $V_{CC}$ | |
| $V_{CC}$ | Low-side supply voltage | 10 | 20 | |
| $T_A$ | Ambient temperature | -40 | 125 | °C |

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## Electrical Characteristics

$V_{BIAS}$ ($V_{CC}$, $V_{BS}$)=15.0 V, $T_A$ = 25°C, unless otherwise specified. The $V_{IN}$ and $I_{IN}$ parameters are referenced to COM. The $V_O$ and $I_O$ parameters are referenced to $V_S$ and COM and are applicable to the respective outputs HO and LO.

| Symbol | Characteristics | Test Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $V_{CCUV+}$ $V_{BSUV+}$ | $V_{CC}$ and $V_{BS}$ supply under-voltage positive going threshold | | 8.2 | 9.2 | 10.0 | V |
| $V_{CCUV-}$ $V_{BSUV-}$ | $V_{CC}$ and $V_{BS}$ supply under-voltage negative going threshold | | 7.6 | 8.7 | 9.6 | |
| $V_{CCUVH}$ $V_{BSUVH}$ | $V_{CC}$ supply under-voltage lockout hysteresis | | | 0.6 | | |
| $I_{LK}$ | Offset supply leakage current | $V_B$=$V_S$=200 V | | | 50 | µA |
| $I_{QBS}$ | Quiescent $V_{BS}$ supply current | $V_{IN}$=0V or 5 V | | 45 | 120 | |
| $I_{QCC}$ | Quiescent $V_{CC}$ supply current | $V_{IN}$=0V or 5 V | | 70 | 180 | |
| $I_{PBS}$ | Operating $V_{BS}$ supply current | $f_{IN}$=20 kHz, rms value | | | 600 | µA |
| $I_{PCC}$ | Operating $V_{CC}$ supply current | $f_{IN}$=20 kHz, rms value | | | 600 | |
| $V_{IH}$ | Logic "1" input voltage | | 2.9 | | | V |
| $V_{IL}$ | Logic "0" input voltage | | | | 0.8 | |
| $V_{OH}$ | High-level output voltage, $V_{BIAS}$-$V_O$ | $I_O$=20 mA | | | 1.0 | |
| $V_{OL}$ | Low-level output voltage, $V_O$ | | | | 0.6 | |
| $I_{IN+}$ | Logic "1" input bias current | $V_{IN}$=5 V | | 10 | 20 | µA |
| $I_{IN-}$ | Logic "0" input bias current | $V_{IN}$=0 V | | 1.0 | 2.0 | |
| $I_{O+}$ | Output high short-circuit pulsed current | $V_O$=0 V, $V_{IN}$=5 V with PW<10 µs | 250 | 350 | | mA |
| $I_{O-}$ | Output low short-circuit pulsed current | $V_O$=15 V, $V_{IN}$=0 V with PW<10 µs | 500 | 650 | | |
| $V_S$ | Allowable negative $V_S$ pin voltage for HIN signal propagation to HO | | | -9.8 | -7.0 | V |

## Dynamic Electrical Characteristics

$V_{BIAS}$ ($V_{CC}$, $V_{BS}$)=15.0 V, $V_S$=COM, $C_L$=1000pF and, $T_A$ = 25°C, unless otherwise specified.

| Symbol | Characteristics | Test Condition | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| $t_{on}$ | Turn-on propagation delay | $V_S$=0 V | 100 | 170 | 300 | ns |
| $t_{off}$ | Turn-off propagation delay | $V_S$=0 V or 200 V[5] | 100 | 200 | 300 | |
| $t_r$ | Turn-on rise time | | 20 | 60 | 140 | |
| $t_f$ | Turn-off fall time | | | 30 | 80 | |
| MT | Delay matching, HS & LS turn-on/off | | | | 50 | |

**Note:**

5. This parameter guaranteed by design.

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## Typical Characteristics

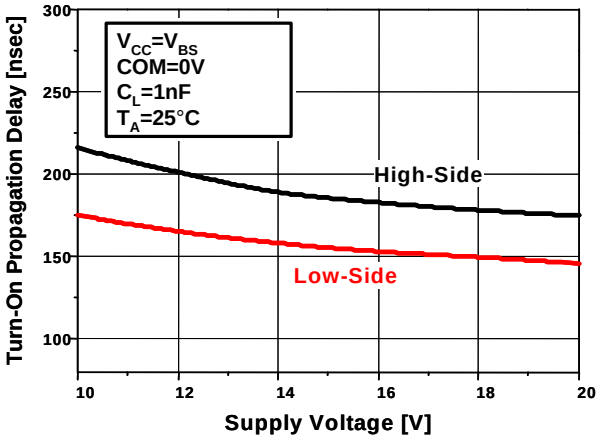



Figure 4. Turn-On Propagation Delay vs. Supply Voltage

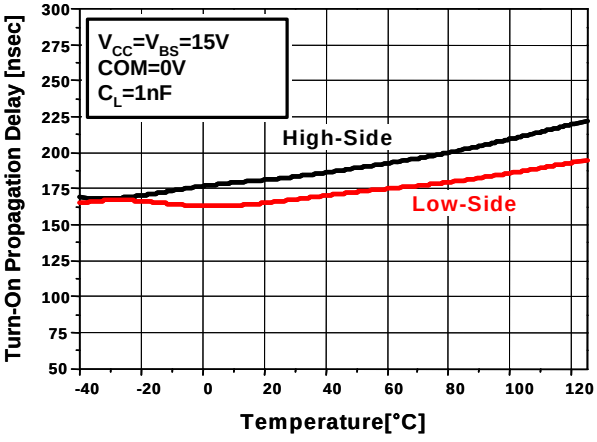



Figure 5. Turn-On Propagation Delay vs. Temp.

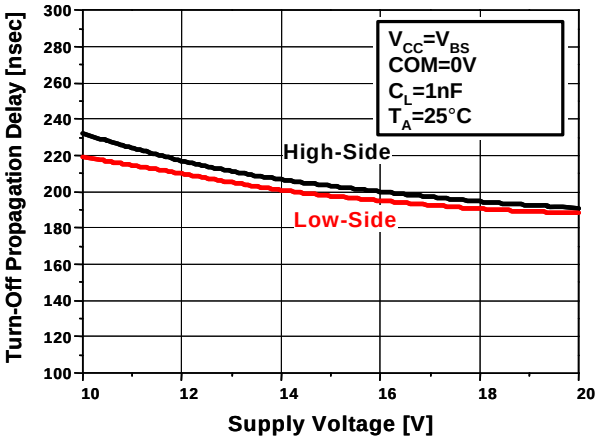



Figure 6. Turn-Off Propagation Delay vs. Supply Voltage

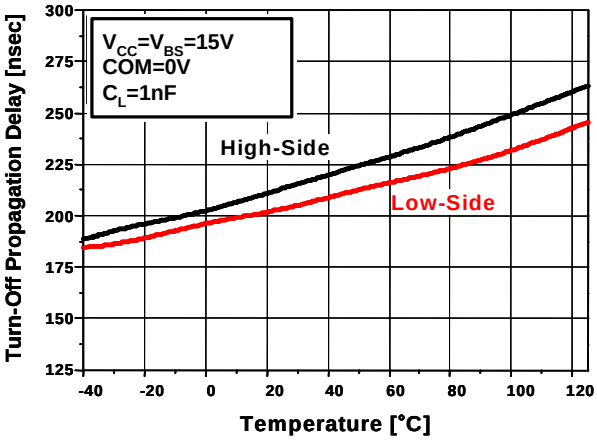



Figure 7. Turn-Off Propagation Delay vs. Temp.

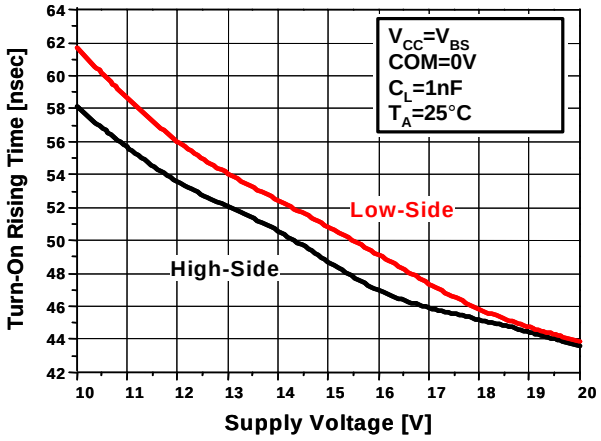



Figure 8. Turn-On Rising Time vs. Supply Voltage

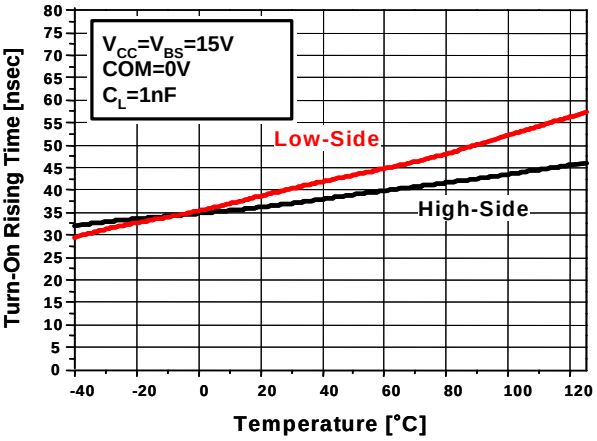



Figure 9. Turn-On Rising Time vs. Temp.

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## Typical Characteristics (Continued)

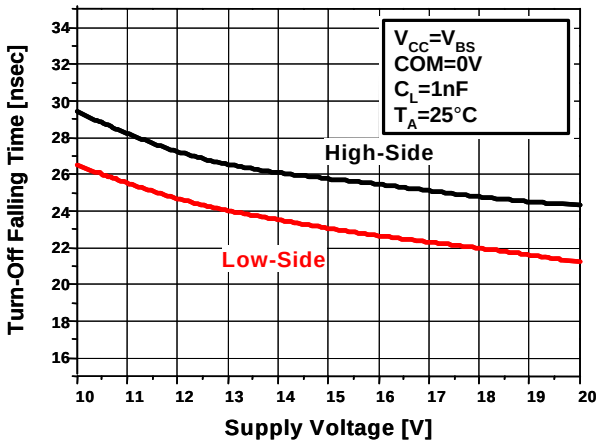



Figure 10. Turn-Off Falling Time vs. Supply Voltage

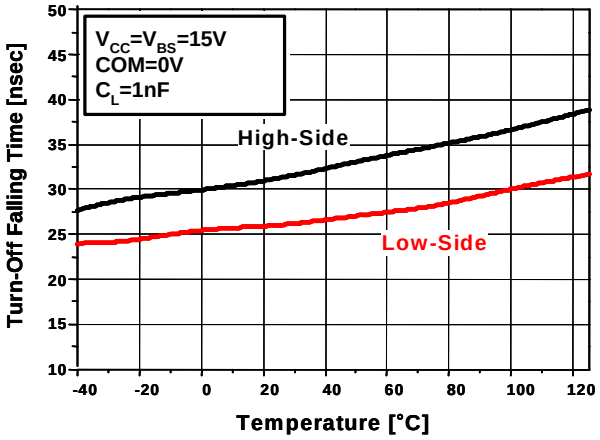



Figure 11. Turn-Off Falling Time vs. Temp.

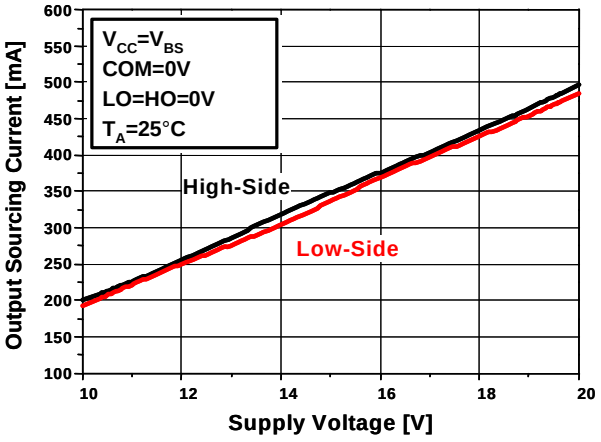



Figure 12. Output Sourcing Current vs. Supply Voltage

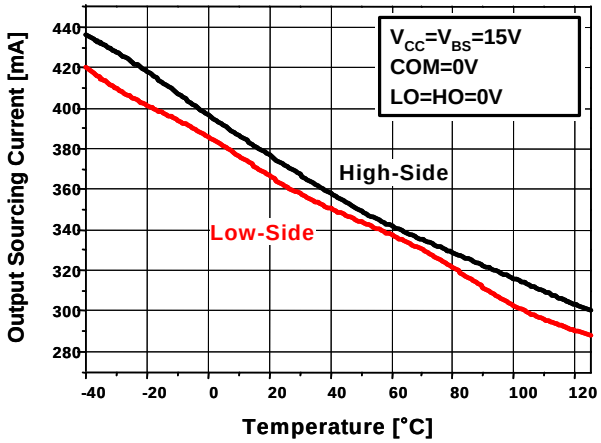



Figure 13. Output Sourcing Current vs. Temp

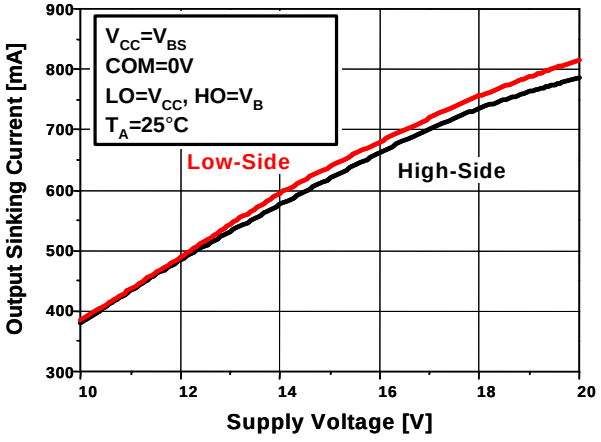



Figure 14. Output Sinking Current vs. Supply Voltage

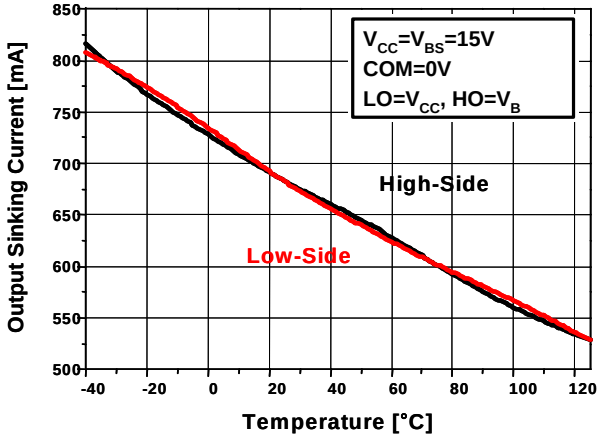



Figure 15. Output Sinking Current vs. Temp.

## Typical Characteristics (Continued)

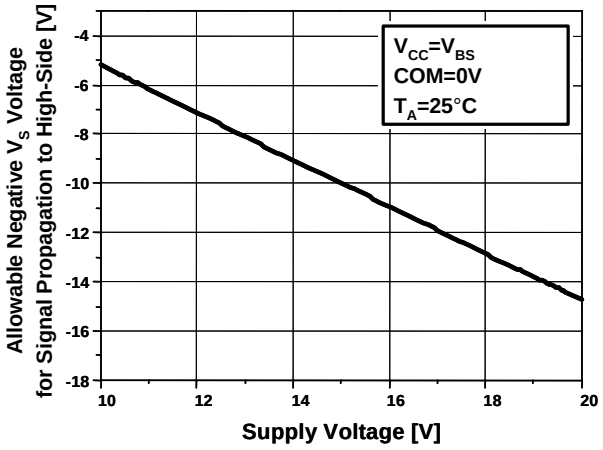



**Figure 16. Allowable Negative $V_S$ Voltage for Signal Propagation to High Side vs. Supply Voltage**

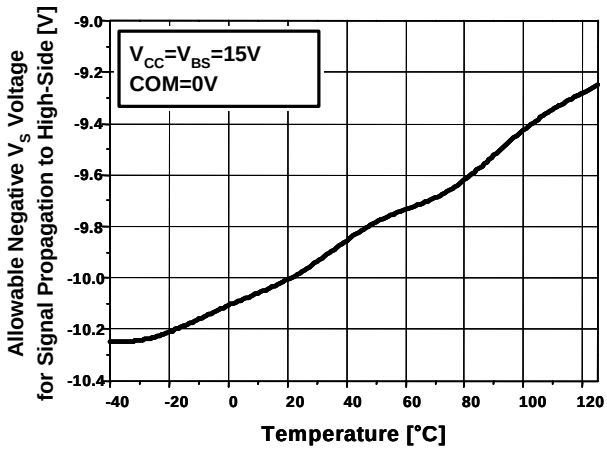



**Figure 17. Allowable Negative $V_S$ Voltage for Signal Propagation to High Side vs. Temp.**

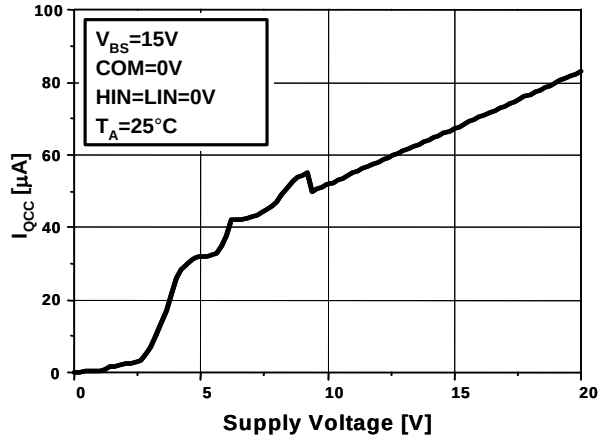



**Figure 18. $I_{QCC}$ vs. Supply Voltage**

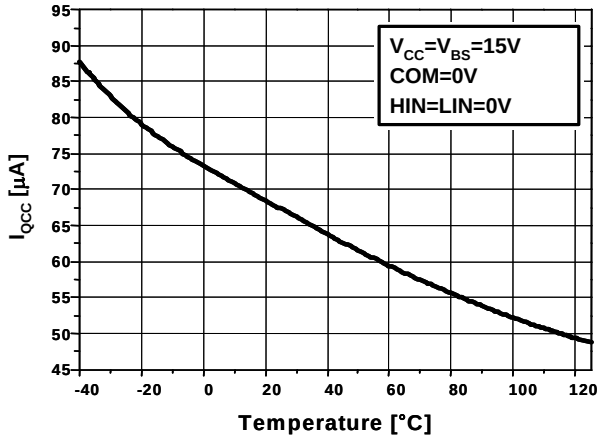



**Figure 19. $I_{QCC}$ vs. Temp.**

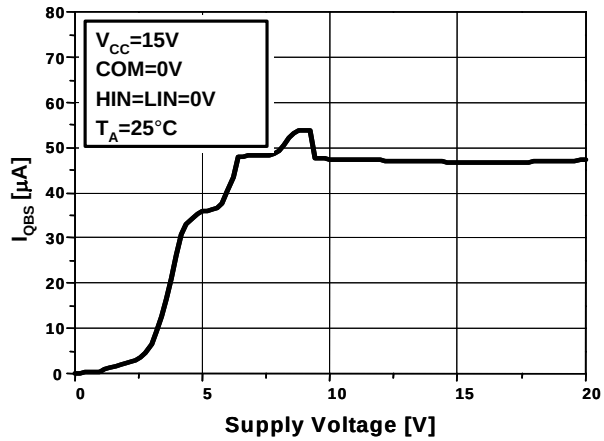



**Figure 20. $I_{QBS}$ vs. Supply Voltage**

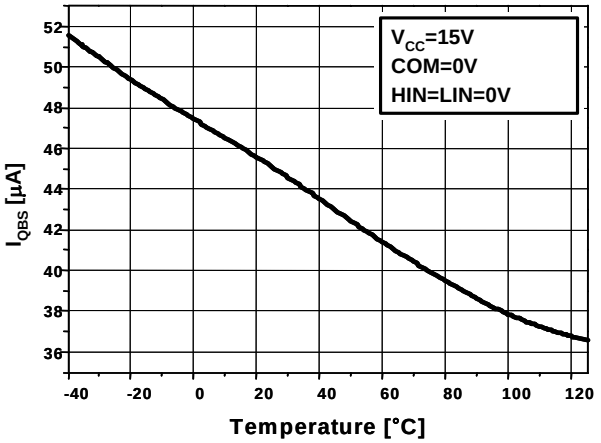



**Figure 21. $I_{QBS}$ vs. Temp.**

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## Typical Characteristics (Continued)

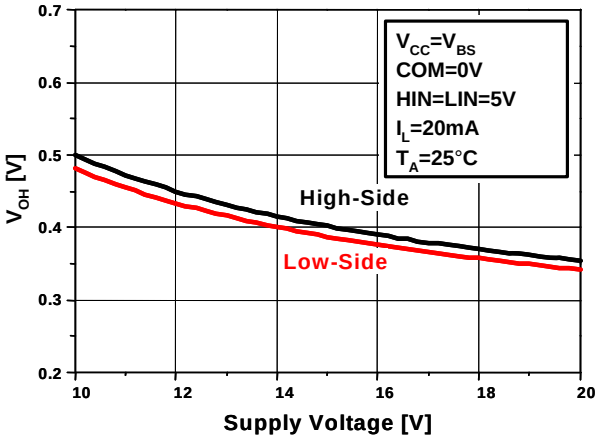



Figure 22. High-Level Output Voltage vs. Supply Voltage

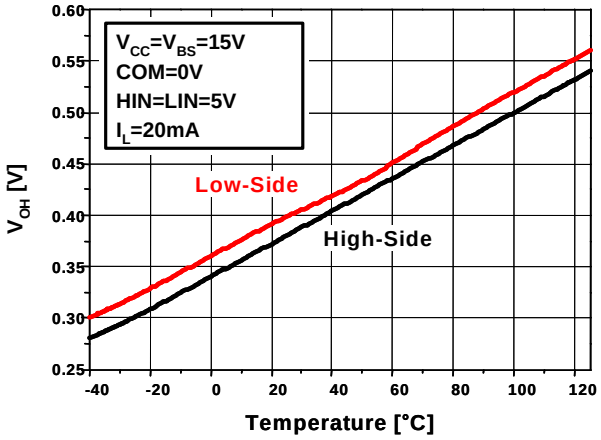



Figure 23. High-Level Output Voltage vs. Temp.

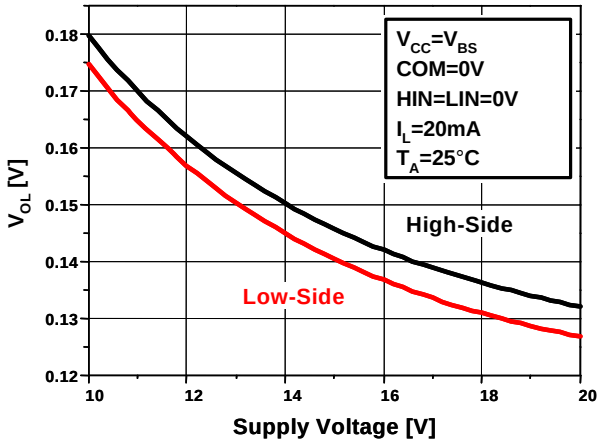



Figure 24. Low-Level Output Voltage vs. Supply Voltage

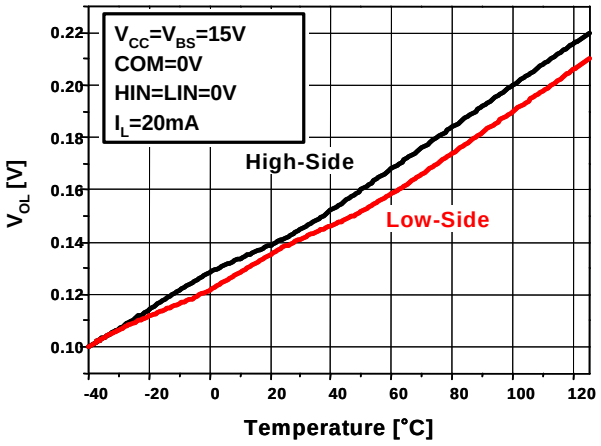



Figure 25. Low-Level Output Voltage vs. Temp.

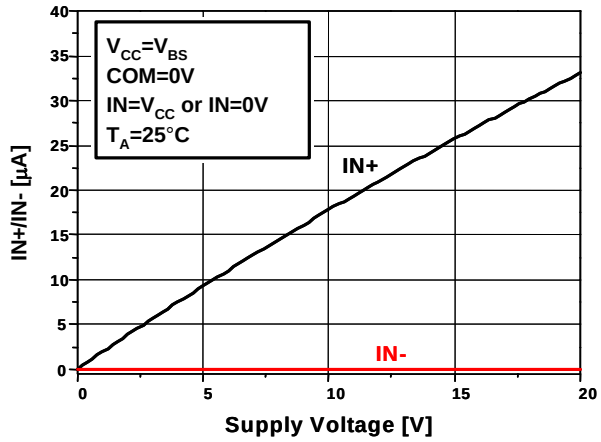



Figure 26. Input Bias Current vs. Supply Voltage

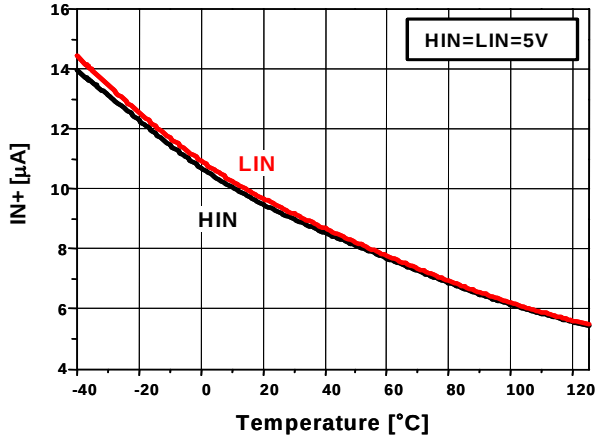



Figure 27. Input Bias Current vs. Temp.

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## Typical Characteristics (Continued)

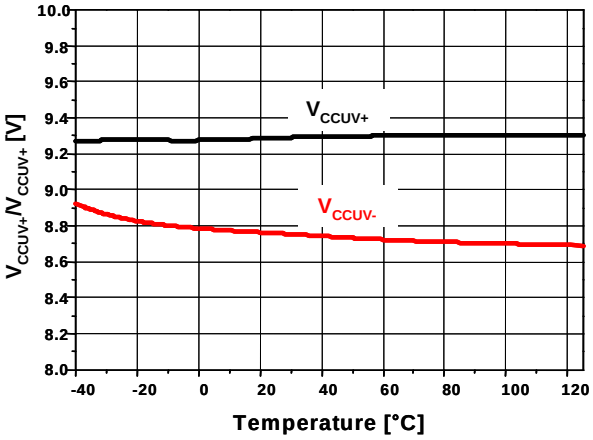



Figure 28. $V_{CC}$ UVLO Threshold Voltage vs. Temp.

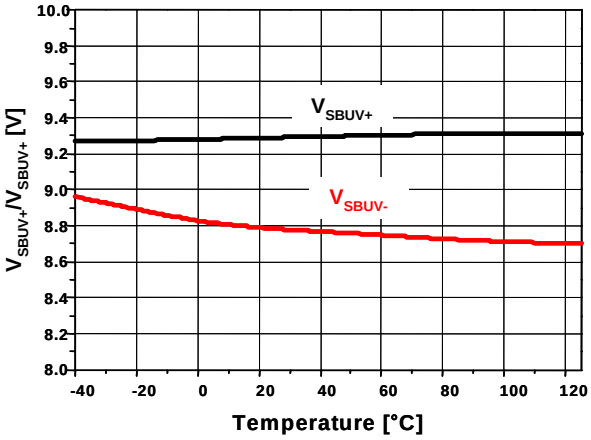



Figure 29. $V_{BS}$ UVLO Threshold Voltage vs. Temp.

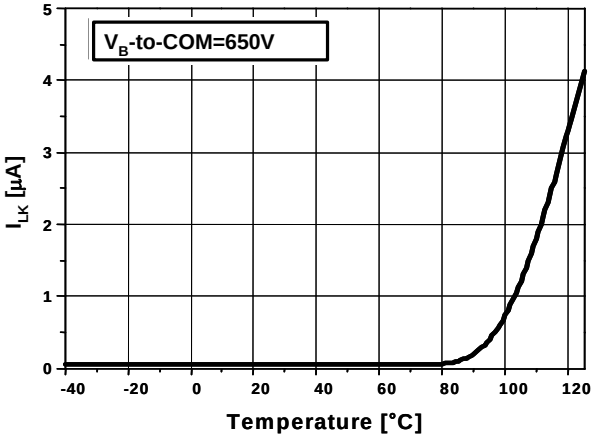



Figure 30. $V_B$ to COM Leakage Current vs. Temp.

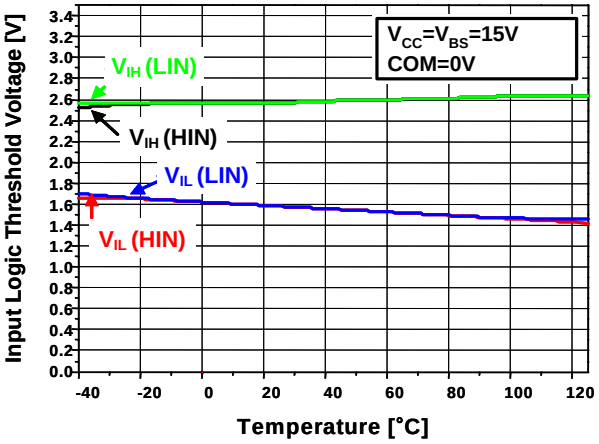



Figure 31. Input Logic Threshold Voltage vs. Temp.

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## Typical Characteristics (Continued)

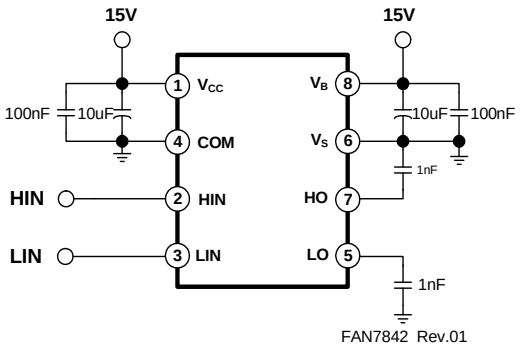



**Figure 32. Switching Time Test Circuit**

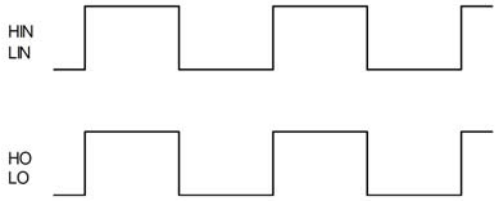



**Figure 33. Input / Output Timing Diagram**

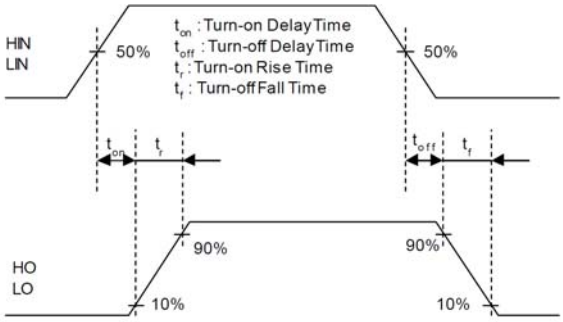



**Figure 34. Switching Time Waveform Definition**

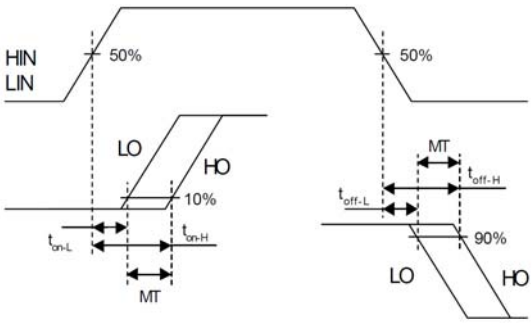
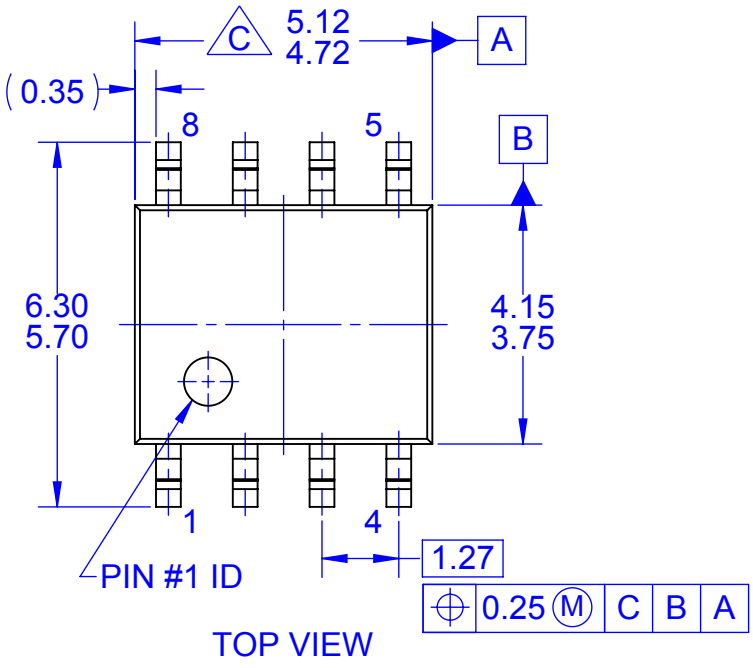
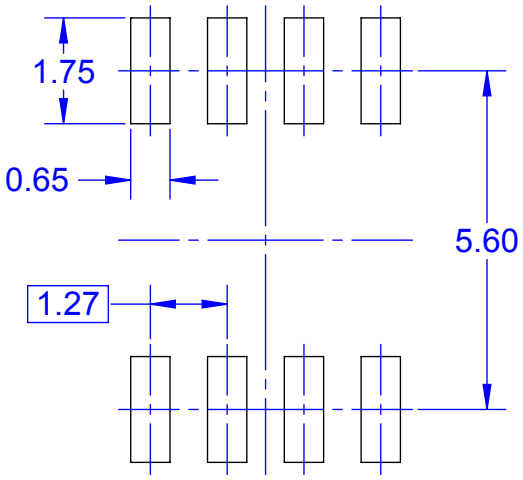


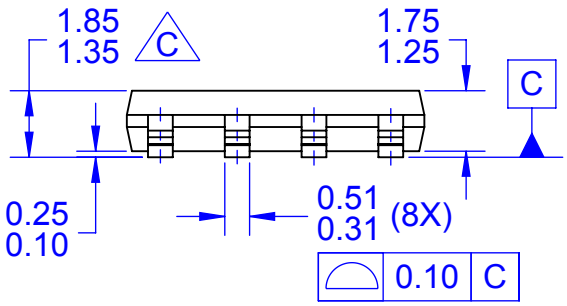

**Figure 35. Delay Matching Waveform Definition**



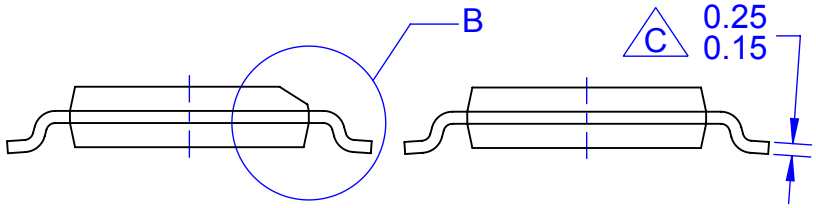

TOP VIEW




LAND PATTERN RECOMMENDATION



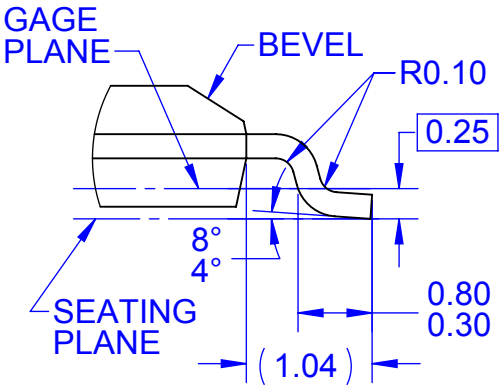

FRONT VIEW




OPTION A
BEVEL EDGE

OPTION B
NON-BEVEL EDGE

SIDE VIEW




DETAIL "B"
SCALE 2:1

NOTES: UNLESS OTHERWISED SPECIFIED

A. THIS PACKAGE CONFORMS TO JEDEC MS-012 VARIATION A EXCEPT WHERE NOTED.
B. ALL DIMENSIONS ARE IN MILLIMETERS

C OUT OF JEDEC STANDARD VALUE

D. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR EXTRUSIONS.
E. LAND PATTERN AS PER IPC SOIC127P600X175-8M
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